TAI NGUYEN PHU

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EDUCATION

University of Information Technology (UIT - VNUHCM) **Sep 2022 - Nov 2025**
Bachelor of Computer Science **GPA:** 3.5/4.0

Le Khiet High School for the Gifted **Sep 2019 - Jun 2022**
Information Technology Major **GPA:** 3.5/4.0

SKILLS

Languages: Proficient in Python, C/C++; Basic in JavaScript, SQL, Bash, HTML, CSS
BE & Deployment: REST APIs (FastAPI, Flask); Git, GitHub Actions, Docker, Weights & Biases
AI/ML: PyTorch, TensorFlow, Pydantic, Pandas, Scikit-learn, NumPy, Chroma, FAISS
NLP/CV: Transformers, PEFT, SFT, SentencePiece, LangChain, LangGraph, OpenCV, YOLO, Detectron2
Courseworks: Linear Algebra, Calculus, Probability & Statistics, Data Structures & Algorithms

PROJECTS

Multimodal Agentic RAG System for PDF documents **Jun 2025 - Aug 2025**

- Designed and implemented an end-to-end Agentic RAG workflow with LangGraph, building modular nodes for generation, retrieval, analysis, rewriting, and synthesis.
- Engineered a multimodal PDF ingestion pipeline leveraging OpenAI Vision and Docling to enable structured extraction and intelligent querying across text and images.
- Integrated Milvus vector store with BM25 hybrid retrieval and BGE-M3 embeddings, orchestrated and monitored via Docker and Attu for scalable deployment.
- Enhanced transparency and multi-turn interaction by implementing conversational memory and detailed agent step tracing within the workflow.
- Delivered a Flask-based web application supporting PDF/image uploads, model selection, real-time workflow visualization, and an interactive chat interface.

Vietnamese Legal Document Semantic Retrieval System **Mar 2025 - Apr 2025**

- Fine-tuned a multilingual SBERT model on a corpus of 100K+ Vietnamese legal documents using MNR and advanced batch sampling strategies, enhancing semantic retrieval performance for domain-specific queries.
- Attained strong benchmark results with NDCG@10 of 60.4% and MAP@10 of 53.6% on MTEB (BKAI Legal Retrieval), demonstrating effectiveness in legal information retrieval tasks.
- Built a complete pipeline encompassing data preprocessing, contrastive fine-tuning, task-specific MTEB evaluation, and deployment workflows.
- Engineered a FAISS-based retrieval system with an interactive Gradio interface, enabling efficient semantic search and user accessibility.
- Containerized the entire system with Docker to ensure reproducibility, portability, and seamless deployment across environments.

Glassdoor 2024 Data Science Job Salary Prediction **Dec 2024 - Jan 2025**

- Crawled 2024 Glassdoor Data Science jobs and performed comprehensive preprocessing, EDA, and feature engineering.
- Achieved $R^2 = 0.82$ using XGBoost, significantly outperforming the Linear Regression baseline ($R^2 = 0.71$).
- Enhanced XGBoost predictive performance and robustness through advanced hyperparameter tuning with Optuna.
- Experimented with LLM-based regression approaches via few-shot prompting with OpenAI's GPT-5 SDK and supervised fine-tuning (SFT + QLoRA) with LLaMA 3.1.
- Benchmarked model performance across traditional ML, few-shot LLMs, and fine-tuned LLMs; built a salary prediction Streamlit app based on the three modeling approaches.

CERTIFICATIONS

Natural Language Processing Specialization
Deep Learning Specialization
Machine Learning Specialization
IELTS Academic | 7.0 *Overall*

ACHIEVEMENTS

Kaggle Competition: "Credit Risk Model Stability" | *Rank* 1096/3858
UIT Global Scholarship